Chapter - 1

LASER

Ques-1 What is Laser? write its characteristics.

Ans- Laser is an acronym for Light Amplification by Stimulated emission of radiation. The first LASER was built by Theodore H Maiman in the year 1960.

Characteristics of a LASER beam –

⇒ The LASER beam has the following four characteristics.

1. LASER beam is highly monochromatic.
2. LASER beam is highly coherent.
3. LASER beam is highly directional.
4. LASER is a high intensity beam of light

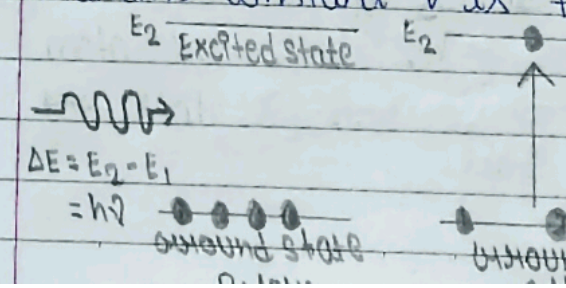
Ques-2 Discuss the interaction of radiation with matter in three possible ways.

Ans- The interaction of radiation with matter :-

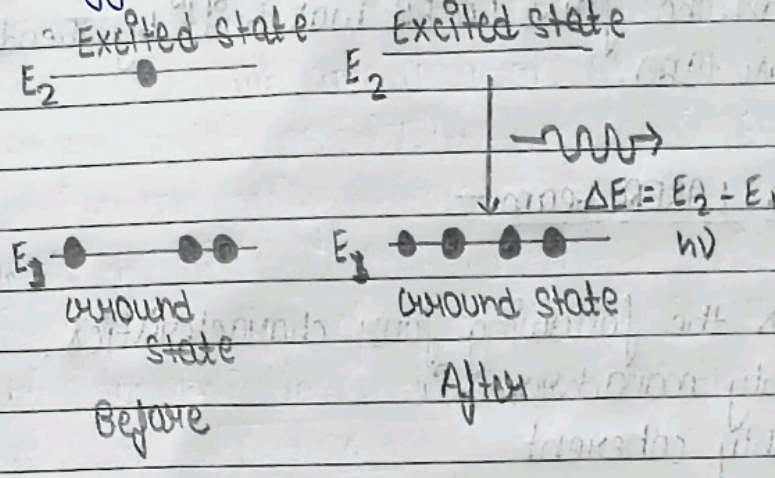
- Induced absorption
- Spontaneous emission
- Stimulated emission

Induced Absorption :- When a photon of light energy is incident on the atom then the photon is absorbed. This process is induce by the photon and hence it is called Induced Absorption.

⇒ Consider an atom in a lower energy states E_1, it will excite to higher energy states E_2 by absorbing the incident photon of energy $E = h\nu = E_2 - E_1$. Here E_1 energy of the lower energy state, E_2 is the energy of the higher energy state, h is the Planck's constant ν is the frequency of photon.

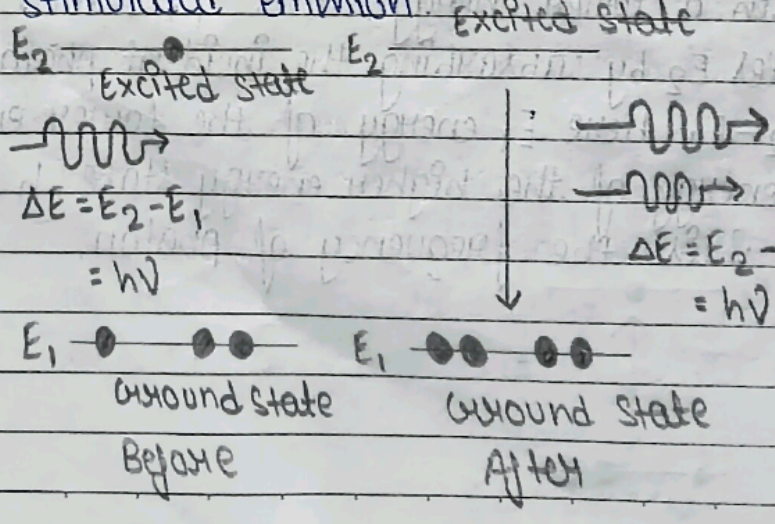


Spontaneous Emission:- It is the process of emission of photon when an atom transits from higher energy level to lower energy level without the influence of any external energy.



⇒ An electron in the higher energy state of an atom makes a transition to lower energy state without the action of any external agency, the energy of the photon emitted is given by $E = h\nu = E_2 - E_1$. In this process the emitted photon need not travel in the same direction. Thus the light beam is not directional.

Stimulated Emission: When a photon of suitable energy interacts with an atom in the higher energy state then the atom is stimulated (forced) to make transition from higher energy state to a lower energy state with the emission of a photon. Both the incident photon and the emitted photons are coherent and travel in the same direction. Thus the process is called stimulated emission.



⇒ when a photon of energy $h\nu = E_2 - E_1$ interacts with an atom in the higher energy state the stimulated emission takes place with the emission of two photons of same energy that are highly directional and coherent. Thus stimulated emission could be used to generate a highly coherent directional beam of light.

3. Explain population inversion, meta stable state & resonant cavity.

Ans - Popular inversion :- If a system is under thermal equilibrium the number of atoms in excited state is less than the no. of atoms in the lower energy state. For the production of LASER number of stimulated emission must be more when compared to induced absorption and spontaneous emission. This is possible only if the number of atoms in the lower energy state and is called population inversion. The means of acheiving population inversion by supplying energy from a suitable source is called Pumping. In addition, to have more stimulated emissions, the life time of atoms in the excited state must be longer. Thus the essential conditions for population inversion are -

1. Higher energy state should posses a longer life time.
2. The number of atoms in the higher energy state must be greater than the number of atoms in the lower energy state.

Meta stable state: The life time of an energy level is of the order of 10^{-8} second. If an atom posses unusual longer lifetime in an energy state such as state is referred to as a meta-stable state. Usually the life time of meta-stable state varies from 10^{-2}s to 10^{-3}s. Population inversion could be achieved with the help of three energy state with one of them a meta-stable state. The population inversion is achieved b/w the state E_2 and E_1 as state E_2 is a meta-stable state.

Resonant cavity:- The LASER cavity consist of an active medium bound between two highly parallel mirrors. The reflection of pho from the mirrors results in multiple transverse of photons through the active medium inducing more and more stimulate emissions. Thus amplification of light is achieved. This also helps to top certain permissible part of LASER Energy from the active medium. The cavity resonates and the output will be maximum when the distance 'l' between the mirrors is equal to an integral multiple of $\lambda/2$ Here λ is the wavelength of incident suitable radiation the length of the LASER cavity is expres as $\boxed{l = \frac{n\lambda}{2}}$

4. Explain the rate of observation & rate of Emission. Derive th expression for energy density using Einstein's coefficient.

Ans - Expression for energy density in terms of Einstein's coefficient: consider 2 energy levels E_1 & E_2 of system of atoms with N_1 & N_2 are population of energy levels respectively. let U be the energy density of incident beam of radiati of frequency ν.
us be the absorption & 2 emission proces.

⇒ The rate of absorption is proportional to $N_1 U$

Rate of absorption = $B_{12} N_1 U_\nu$ — (1)

B_{12} is the proportionality constant called Einstein coefficient of induced absorption.

⇒ Spontaneous emission depends on N_2 & independent of energy density.

The rate of spontaneous emission = $A_{21} N_2$ — (2)

⇒ The rate of stimulated emission is directly proportion to $N_2 U_\nu$.

⇒ The rate of stimulated emission = $B_{21} N_2 U_\nu$ — (3)

Rate of absorption = (Rate of spontaneous emission + Rate of stimulated emission)

$$B_{12}N_1U_V = A_{21}N_2 + B_{21}N_2U_V$$

$$U_V(B_{12}N_1 - B_{21}N_2) = A_{21}N_2$$

$$U_V = \frac{A_{21}N_2}{(B_{12}N_1 - B_{21}N_2)}$$

$$U_V = \frac{A_{21}}{B_{21}}\left[\frac{N_2}{\left(\frac{B_{12}}{B_{21}}N_1 - N_2\right)}\right]$$

$$U_V = \frac{A_{21}}{B_{21}}\left[\frac{1}{\left(\frac{B_{12}N_1}{B_{21}N_2}\right) - 1}\right] \quad \text{---(4)}$$

By Boltzmann's law, $N_2 = N_1 e^{-\left(\frac{E_2 - E_1}{kT}\right)}$

$$N_1/N_2 = e^{h\nu/kT}$$

Eqn (4) becomes $U_V = \frac{A_{21}}{B_{21}}\left[\frac{1}{\left(\frac{B_{12}}{B_{21}}e^{(h\nu/kT)} - 1\right)}\right]$ ---(5)

By Planck's law, $U_V = \frac{8\pi h\nu^3}{c^3}\left[\frac{1}{\left(e^{(h\nu/kT)} - 1\right)}\right]$ ---(6)

comparing eqn (5) & (6)

$$\frac{A_{21}}{B_{21}} = 8\pi h\nu^3/c^3 \quad \& \quad \frac{B_{12}}{B_{21}} = 1 \text{ i.e } B_{12} = B_{21}$$

The probability of induced absorption is equal to the stimulated emission.

Therefore, A_{12} is written as A & B_{12}, B_{21} written as B.

Eqn (5) becomes

$$U_V = \frac{A}{B}\left[\frac{1}{(e^{(h\nu/kT)} - 1}\right]$$

Rate of Induced absorption: The rate of induced absorption is defined as the no. of induced absorption per second per unit

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volume in unit time rate of absorption depends on 1 No. of atoms in the lower energy state N_1 2. The incident energy density E_v.

Hence, 1) Rate of induced absorption $\propto N_1 E_v$

2) Rate of induced absorption $= B_{12} N_1 E_v$.

Here B_{12} is proportionality constant called einsteins coefficie of induced absorption.

Rate of Spontaneous emission :- The no. of spontaneous emissio per unit volume in unit time called rate of emission. Rate of spontaneous emission depends on 01) Rate of spontaneous emission $\propto N_2$. 02) Rate of spontaneous emission $= A_2 N_2$ Here A_{21} is this proportionality constant called einstein's coefficient of spontaneous emission

Rate of stimulated Emission :- The no. of stimulated emiss per unit volume in unit time is called rate of stimulated emission, rate of stimulated emission depends upon.

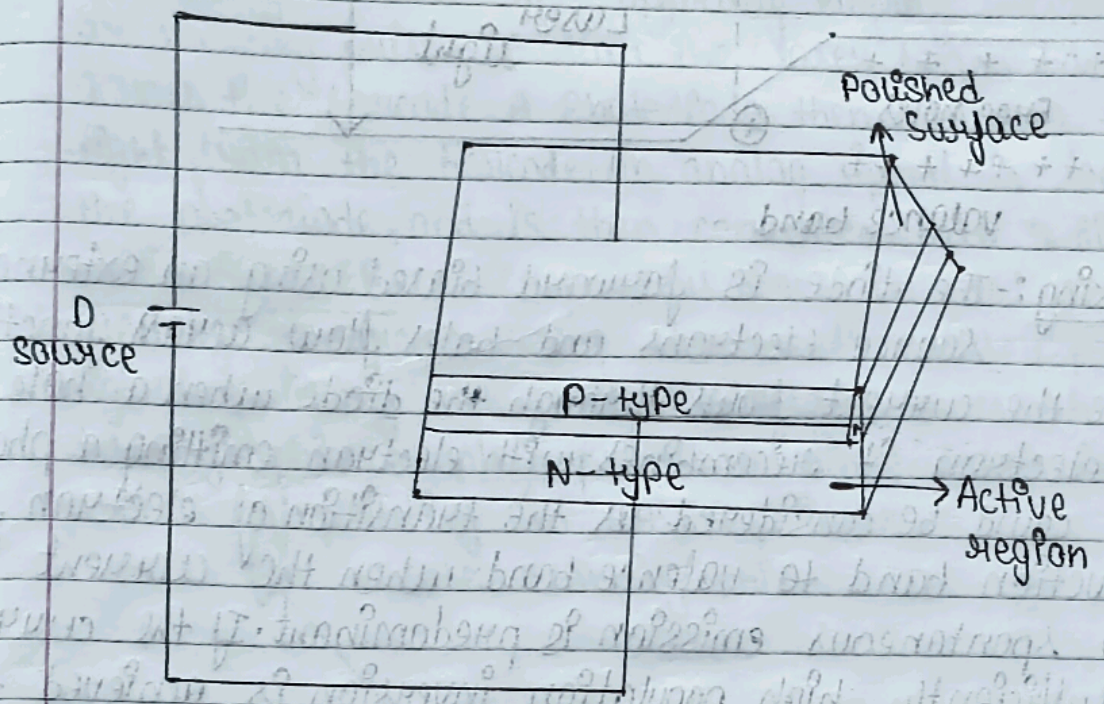
1. No. of atoms in the higher energy state (N_2).
2. The energy density (E_v)

Hence,

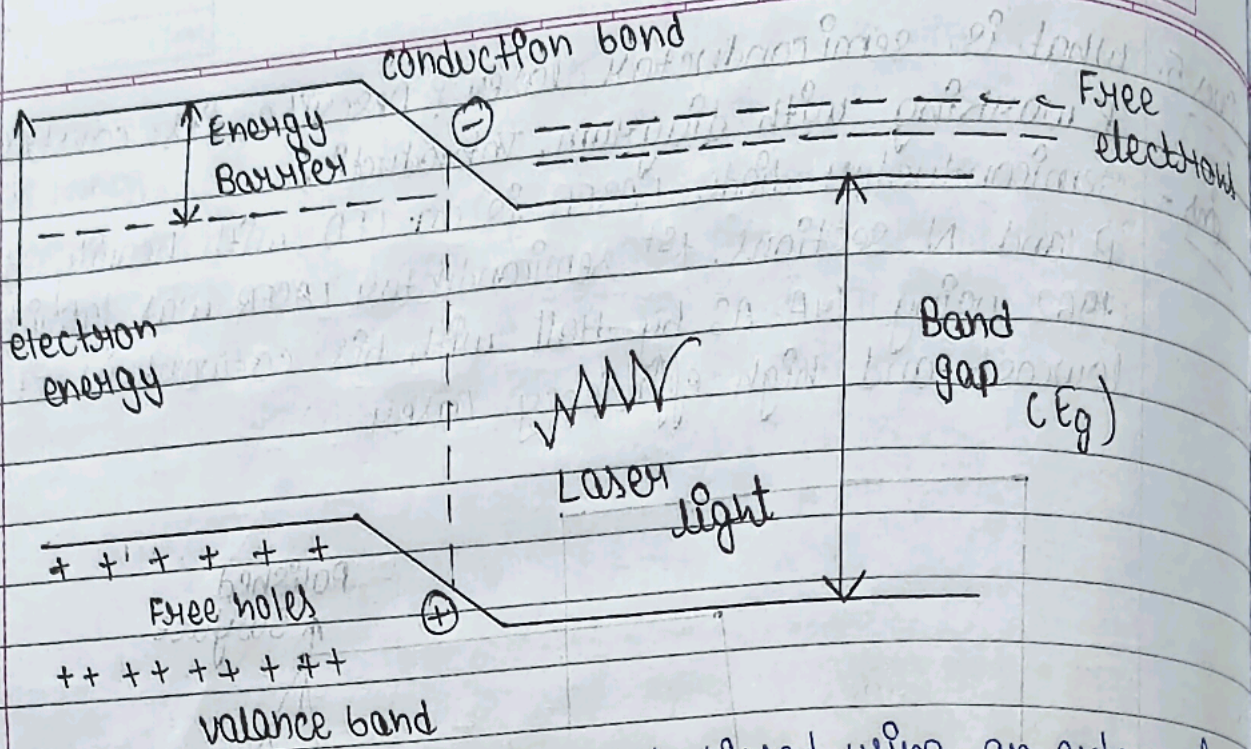
1. The rate of stimulated emission $\propto N_2 E_v$.
2. Rate of stimulated emission $= B_{21} N_2 E_v$

Ans-5. What is semiconductor laser? Describe its constructing & working with diagram. Introduction

Ans- Semiconductor diode LASER is an LED with heavily doped P and N sections. 1st semiconductor LASER was fabricated in 1962 using GaA-AS by Hall with his co-workers. It is a lowcost and high efficiency laser.



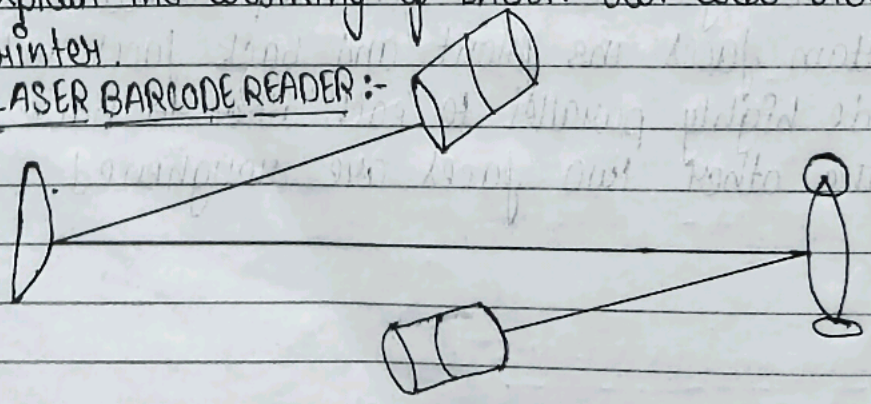
Construction:- The GaA-AS LASER diode belongs to direct band gap semiconductors. The n-section is derived by doping the substrate with tellurium and P-Section is derived by doping the substance with Zn the diode used is in the form of cube with dimension 0.4mm the depletion region is of thickness 0.1NM & lies horizontal as shown fig. The current is passed through the ohmic contacts provided to the top and bottom faces. The front and back faces are polished and made highly parallel to each other to have a LASER cavity. The other two faces are roughened.



Working:- The diode is forward biased using an external source. Electrons and holes flow across function. Hence the current flows through the diode when a hole meet an electron it recombines with electron emitting a photon. this could be considered as the transition of electron from conduction band to valence band when the current is low spontaneous emission is predominant. If the current is sufficiently high population inversion is achieved the photons liberated initially due to spontaneous emission induced further stimulated emissions, the laser cavity helps in the amplication of light finally this results in an avalanche of photons and hence the LASER action emitted is 840nm.

6. Explain the working of LASER bar code reader & LASER printer.

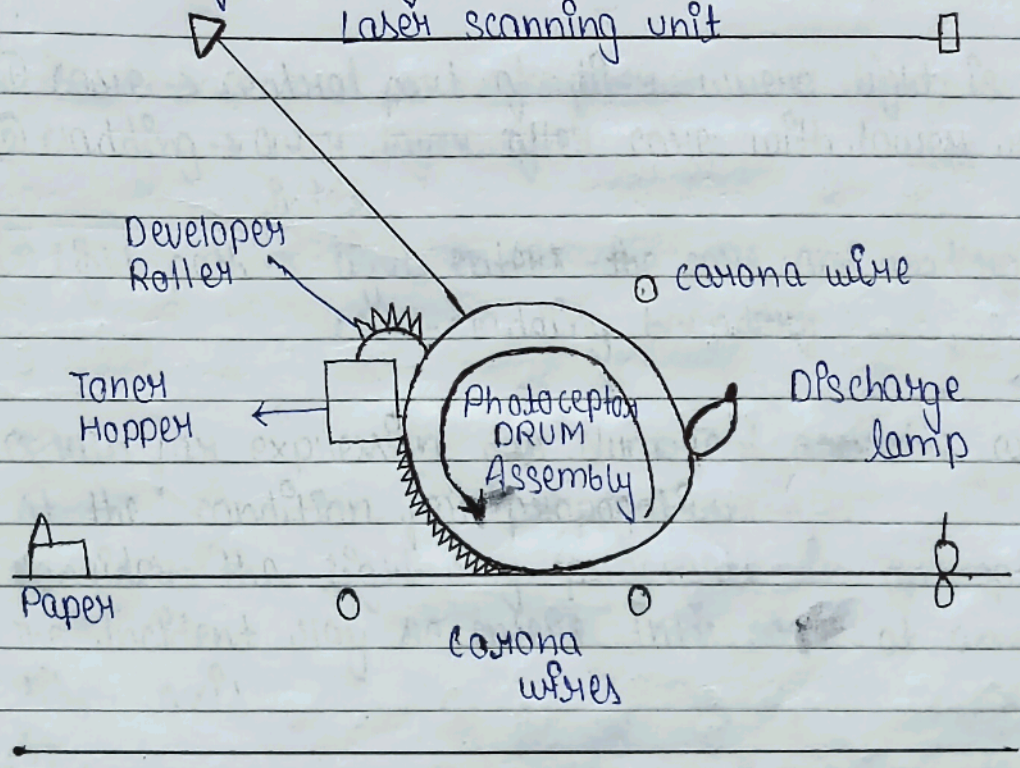
Ans - LASER BARCODE READER :-



⇒ A barcode is a printed series of parallel bars or lines of varying width that is used for entering data into a computer system.

⇒ A barcode scanner/reader is a device with lights and a sensor that decodes and captures the information contained in barcodes. laser scanners use a laser beam as a light source and typically employ oscillating mirrors or rotating prism to scan the laser beam back & forth across the barcode. A photodiode then measures the reflected light from the barcode. An analog signal is located from the photodiode, and is then converted into a digital signal.

LASER PRINTER:- LASER printers were invented at XEROX in 1969 by researcher Gary starkweather laser printers are digital printing devices that are used to create high quality texts graphics on plain printer A diode laser is used in the process of printing in LASER printer.



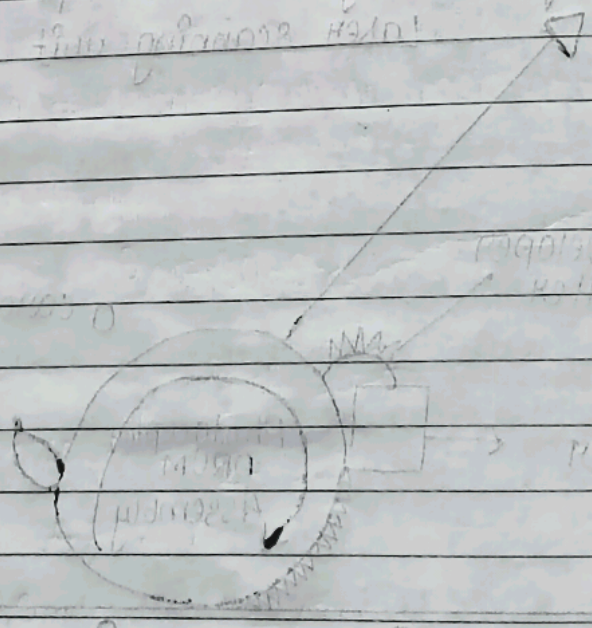
working principle :- A laser bean projects an image of the page to be printed onto an electrically charged rotation photosensitive drum coated with selenium.

⇒ Photo conductivity allows charge to leak away from the

areas which are exposed to light and the area gets positively charged.

⇒ Toner particles are then electrostatically picked up by the drums charged areas which have been exposed to light.

⇒ The drum then prints the image onto paper by direct contact and heat which fuses the ink to the paper.



Chapter-2

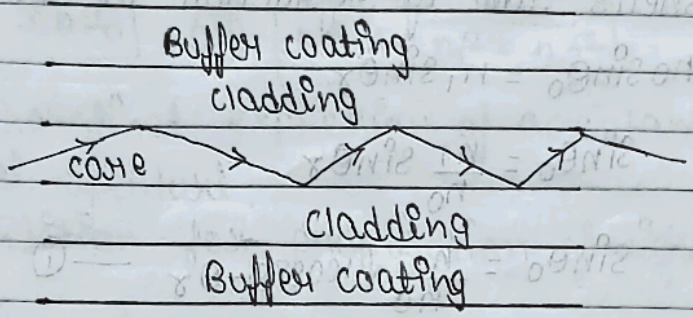
OPTICAL FIBERS

1. what is optical fiber? Explain the principle of O.F.

Ans- optical fibers are the wires and strands made of transparent dielectrics which guide light over longer distances using the phenomenon of Total Internal Refraction

Principle:-

when light enters the core at an angle greater than the critical angle, it undergoes total internal reflection at the core-cladding interface, bouncing back into the core.



① core → central part of fiber where light is transmitted.
② cladding → outer layer after core with lower refractive index.
③ Light path → light enters the core and undergoes TIR at core-cladding boundary.

② Obtain for expression for Numerical aperture and derive at the condition for propogation.

Ans- Consider the light ray propogate in an optical fiber.

⇒ The incident ray AO enters into core at an angle θ_0 to fiber axis.

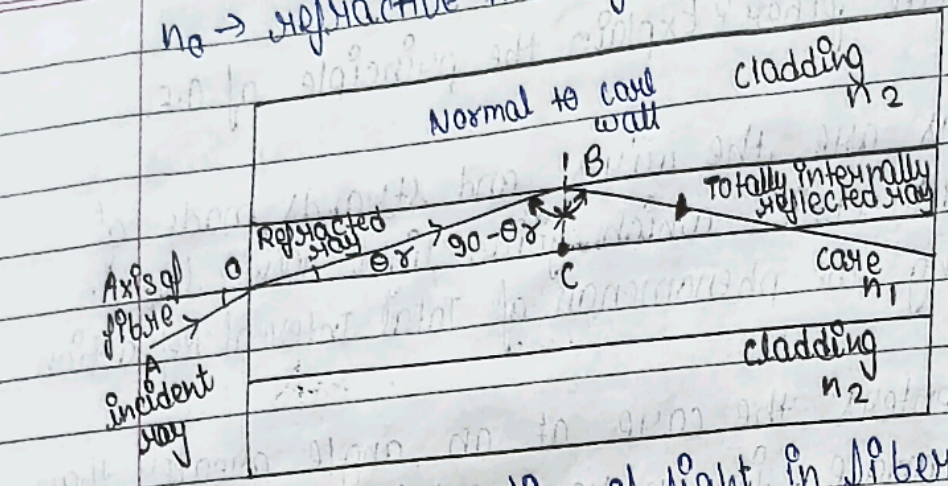
⇒ Let,

n_1, n_2 and n_0 be the refractive indices of the core, cladding and surrounding.

n_1 → refractive index of core

n_2 → refractive index of cladding.

n_0 → refractive index of surrounding.



→ Propagation of light in fiber

Applying Snell's law of refraction at the point of O we h

$$n_0 \sin\theta_0 = n_1 \sin\theta_\gamma$$

$$\sin\theta_0 = \frac{n_1}{n_0} \sin\theta_\gamma$$

$$\sin\theta_0 = \frac{n_1}{n_0} \sqrt{\cos^2\theta - \theta_\gamma} \quad \text{— (1)}$$

At the point B on the interface of core and cladding.

Angle of incidence,

$$\theta_c = 90 - \theta_\gamma$$

Applying Snell's law of refraction at the point B we ha

$$n_1 \sin(90-\theta_\gamma) = n_2 \sin 90^\circ$$

$$n_1 \cos\theta_\gamma = n_2$$

$$\cos\theta_\gamma = \frac{n_2}{n_1} \quad \text{— (2)}$$

Substituting eq^n (2) in eq^n (1) we have

$$\sin\theta_0 = \frac{n_1}{n_0} \left(\sqrt{1.} - \frac{n_2^2}{n_1^2}\right.$$

$$\sin\theta_0 = \frac{n_1}{n_0} \sqrt{\frac{n_1^2 - n_2^2}{n_1}}$$

$$\sin\theta_0 = \frac{n_1}{n_0 n_1}\sqrt{\frac{n_1^2 - n_2^2}{1}}$$

$$\sin\theta_0 = \sqrt{\frac{n_1^2 - n_2^2}{n_0}}$$

the medium surrounding the fiber is air, then $n_0 = 1$

$$\sin\theta_0 = \sqrt{n_1^2 - n_2^2}$$

$$\theta_0 = \sin^{-1}\sqrt{n_1^2 - n_2^2} \quad \text{---(3)}$$

$$\boxed{\sin\theta_0 = \sqrt{n_1^2 - n_2^2} = N.A}$$

$$\boxed{NA = \sin\theta_0} \quad (or) \quad \boxed{NA = \sqrt{n_1^2 - n_2^2}}$$

This is the final expression of acceptance angle and numerical aperture.

⇒ The condition for propogation of light within the fiber is

$$\sin\theta_i < NA$$

NOTE → Acceptance angle :- The maximum angle at (or) below which a ray of light can enter through one end of the fibre still be total internal reflection is called as acceptance angle. The cone is reffered as acceptance cone.

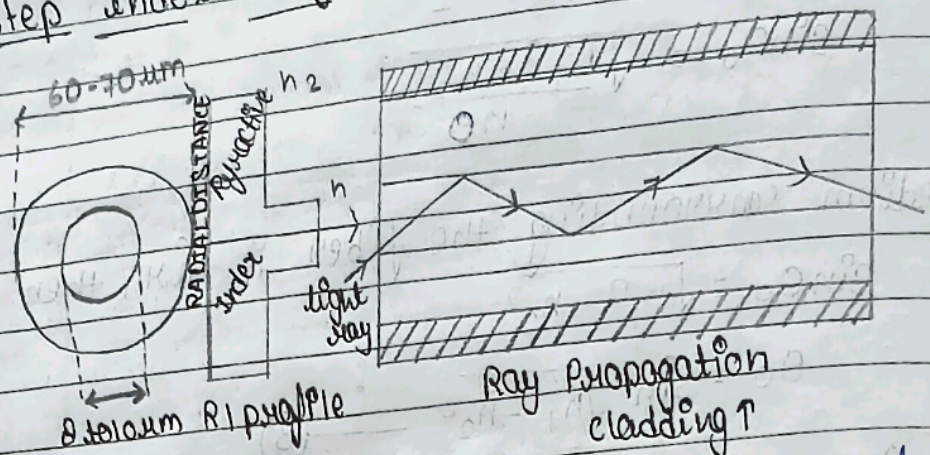
$$\theta_0 = \sin^{-1}\sqrt{n_1^2 - n_2^2}$$

3. Discuss the types of optical fibers based on modes of propogation and RI profile.

Ans- Types of optical fibers are 3, those are on modes of propagation.

① Step index Single-mode fiber
② Step index multi-mode fiber
③ Graded index multi-mode fiber.

① Step index single-mode fiber.



⇒ Here, this consists of very fine thin core of uniform RI surrounded by cladding at the interface it is called step index fiber.
Since, core size small. N.A is also small & support single mode.
⇒ They accept light from LASER source.
* Splicing is diffult. They are used in submarin cables.

② Step index multi-mode fiber:

* This is similar to (SI SMF) with the exception that it has a large core diameter is very large.
* Numerical aperture is large because of large core si and thus support multi-modes.
* They accept light from both LASER as well as from

* They are used in data links

④ Graded Index multi-mode fiber.

⇒ A multi-mode fiber has concentric layers of RI is called GRIN fiber.
⇒ It means RI of the core varies with distance from fiber axis.
⇒ RI is maximum at center & decrease with radial distance towards to core-cladding interface.
⇒ In GRIN fiber acceptance angle & NA diminish with radial distance.
⇒ They accept light from both LASER as well as from LED.
⇒ Used for medium distance communication for ex-telephone link between central faces.

④ Define Attenation. Explain the three types of attention in optical fiber.

Ans- Attenation:- The optical energy (signals) passing through the optical fiber gets reduced progressively. This is due to attenuation. It is called the fiber Loss (or) significant loss

⇒ Three types of Attenation are:-

⇒ Absorption loss:- Loss of signal power occurs due to

absorption of photons associated with the signal.

* Absorption loss is wavelength dependent:

2 types are there

① Extrinsic absorption:- This type of loss in optical fiber is due to absorption of light by the impurities such as OH^- ions & transition metal ions.

Such as Fe^{2+}, Cr^{2+}, Co, Cu

② Intrinsic absorption:- Intrinsic loss in fiber is due to the absorption of light by the material of the fiber glass itself.

* These are insignificant.

⇒ Scattering loss:- Light travelling through the core can get scattered by impurities or small regions with sudden change in RI.

* Rayleigh scattering varies as $\alpha = \frac{1}{\lambda^4}$

Leads to significant Power loss

* Scattering results in loss of photons

* Rayleigh scattering is responsible for maximum loss in optical fibers.

③ Geometric effects:- These occur due to manufacturing defects like irregularities in fiber dimensions during drawing process, during coating, cabling & insulation proces

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⑤ With the help of block diagram, explain point to point communication using optical fiber.

Ans

→ In an optical fiber communication system, the input signals (audio, video or other digital data) are used to modulate light from a source like a LED or a semiconductor LASER is transmitted through optical fiber.

⇒ At the receiving end the signal is demodulates to reproduce the input signal.

⇒ If data transfer takes place between only two devices then, it is called point to point communication.

Steps are - ① 1st voice converted into electrical signal using a transducer.

② It is digitized using a coder.

③ The digitized signal, carries voice information, is fed to optical transmitter.

④ Light source is optical transmitter (LED or LASER diode emits modulated light, transmitted through optical fiber.

⑤ At other end modulated light signal is detected by photo detector & decoded using a decoder.

⑥ Finally the information is converted into analog electrical signal & is fed to a bud speaker, which converts the signal to voice (sound).

6. With a neat diagram, explain fiber optic networking.

Ans:-

Local Area Network:- A local area network (LAN) is a type of computer network that interconnects multiple-computers and computer driven devices in a particular physical location.

* Traditionally copper coaxial cables are used for LAN.

Passive optical LAN:-

⇒ A passive optical network refers to a fiber-optic network utilizing a point to multipoint topology & optical splitters to deliver data from a single transmission point to multiple user endpoints.

* Passive op LANS are built entirely using optical fiber cables.

* These works on a concept of optical network terminals (ONT) & passive optical splinters.

Problem:

1. The refractive indexes the core uncladding are $n_1 = 1.50$ and $n_2 = 1.48$ find the numerical aperture.

Sol^n -

$$N.A = \sqrt{n_1^2 - n_2^2} = \sin\theta_0$$

$$\sin\theta_0 \Rightarrow \theta_0 = \sin^{-1}(N.A)$$

$$N.A = \sqrt{(1.50)^2 - (1.48)^2}$$

$$N.A = \sqrt{2.25 - 2.19}$$

$$N.A = \sqrt{0.0596}$$

$$N.A = 0.24413$$

$$\theta_0 = \sin^{-1}(0.24413)$$

$$\theta_0 = 14.13^0$$

2. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is kept in medium of refractive index 1.33.

Sol^n -

$$\sin\theta_0 = \sqrt{n_1^2 - n_2^2}$$

$$30^\circ = \sqrt{n_1^2 - n_2^2}$$

$$0.5 = n_1 - n_2$$

$$n_2 = n_1 - 0.5$$

$$n_2 = 1.33 - 0.5$$

$$n_2 = 0.83$$

$$\theta_0 = \sin^{-1}(\sqrt{n_1^2 - n_2^2})$$

$$\theta_0 = \sin^{-1}(\sqrt{1.7689 - 0.6889})$$

$$\theta_0 = \sin^{-1}\sqrt{1.08}$$

$$\theta_0 = \sin^{-1}\sqrt{1}$$

$$\theta_0 = 90^\circ$$

③ An optical fiber core refractive index 1.5 & cladding is 3% less than that of core include calculate its Numerical aperture, angle of acceptance, internal critical angle.

Soln. $n_1 = 1.5$

$n_2 = 3\%$ less than that of core $= 1.5 - 3\% = 1.5 - 0.03 = 1.47$

$\sin\theta_0 = \sqrt{n_1^2 - n_2^2}$

$= 0.29$

$\theta_0 = \sin^{-1}(\sqrt{n_1^2 - n_2^2})$

$= 16.8^\circ / 17.36^\circ$

$\theta_0 = \sin^{-1}\left(\frac{n_2}{n_1}\right)$

$\theta_0 = 78.52^\circ$

④ Find an angle attenation of an optical fiber of length when a light signal of power 100mW emerges out of the fiber with the power of 90mW.

Soln. Given,

$L = 500\,m$

$P_{in} = 100\,mW$

$P_{out} = 90\,mW$

$\alpha = ?$

$\alpha = \frac{-10}{L}\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$ in dB/km

$\alpha = \frac{-10}{500}\log_{10}\left(\frac{90}{100}\right)$

$\alpha = \frac{-1}{50}\log_{10}\left(\frac{9}{10}\right)$

$\alpha = 0.02\log_{10}$

$\alpha = -0.02 \times -0.045$

$\alpha = 9 \times 10^{-4}\,dB/km$

⑤ An optical signal propagating in a fiber retain 85% of input power after travelling the distance of 500m, In the fiber. calculating the attenation coefficient.

Sol- Given,

$L = 500m$

$P_{in} = 100\%$.

$P_{out} = 85\%$.

$\alpha = ?$

$$\alpha = \frac{-10}{L} \log_{10}\left(\frac{P_{out}}{P_{in}}\right) \text{ in dB/km}$$

$$\alpha = -0.02 \times -0.07058$$

$$\alpha = 1.41 \times 10^{-3} \text{ dB/km}$$

⑥ The attenation in an optical fiber is 3.6 dB/km. what fraction of it's initial intensity remains after (i) at 1km (ii) After 3 km.

Soln- Given → $L = ?$

$P_{in} = ?$

$P_{out} = ?$

$\alpha = 3.6$ dB/km

1) at 01 km

2) at 03 km

FOR $\alpha = \frac{-10}{L} \log_{10}\left(\frac{P_{out}}{P_{in}}\right)$ in dB/km

$$= \frac{-\alpha L}{10} = \log_{10}\left(\frac{P_{out}}{P_{in}}\right)$$

$$\boxed{\frac{P_{out}}{P_{in}} = 10^{\frac{-(\alpha L}{10})}}$$

(i) at 1km $= \frac{P_{out}}{P_{in}} = 0.43$

(ii) at 3km $= \frac{P_{out}}{P_{in}} = 0.083$

⑦ The Attenation in an optical fiber is 1.5 DB/Km. what is the total length of the optical fiber. If the fraction of the pow is 0.936

Sol:-

$\alpha = 1.5 DB/Km$

$L = ?$

$\frac{P_{out}}{P_{in}} = 0.936$

$\alpha = \frac{-10}{L} \log_{10}\left(\frac{P_{out}}{P_{in}}\right)$

$L = \frac{-10}{1.5} \log_{10}(0.936)$

$L = -0.028 \times 1.5$

$L = 0.191 Km$

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⇒ Derivation of one dimensional schrondinge wave fn equation (Time independent) :-

⇒ A moving particle with velocity $v \rightarrow$ with mass $m \rightarrow$

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad -(1)$$

consider,

$$\boxed{\psi = Ae^{i(\omega t - kx)}} \quad -(2)$$

$\Rightarrow e^{-i(\omega t - kx)}$

$\Rightarrow e^{-i\omega t} \cdot e^{+ikx}$

$$\psi = Ae^{ikx} \longrightarrow \text{space dependent}$$

$$\psi = Ae^{i\omega t} \quad -(3) \longrightarrow \text{Time dependent}$$

$$\psi = Ae^{i\omega t} \psi \quad -(3)$$

diff wrt x

$$\frac{d\psi}{dx} = Ae^{i\omega t} \frac{d\psi}{dx}$$

$$\frac{d^2\psi}{dx^2} = e^{i\omega t} \frac{d^2\psi}{dx^2} \quad -(4)$$

$\Rightarrow \psi = Ae^{i(kx - \omega t)}$

$\psi = e^{i(kx - \omega t)} \psi$

$\psi = e^{-i\omega t} \psi \quad -(3)$

$\frac{d^2\psi}{dy^2} = e^{-i\omega t} \frac{d^2\psi}{dx^2} \quad -(4)$

$Ae^{-i(\omega t - kx)}$

$\frac{d\psi}{dt} = (-i\omega) e^{-i\omega t} \frac{d\psi}{dt}$

$\frac{d^2\psi}{dt^2} = (-i\omega)(i\omega) e^{-i\omega t} \frac{d^2\psi}{dt^2}$

$= -\omega^2 e^{i\omega t} \frac{d^2\psi}{dt^2} \quad -(5)$

$y = A\sin(-\omega t + kx)$

$\frac{d^2y}{dx^2} = \frac{1}{V^2} \cdot \frac{d^2y}{dt^2}$

MODULE-02 Quantum Mechanics

Dual nature of light: The evidence supporting the particle nature of light is provided by the concepts of the photoelectric effect and the compton effect. In contrast, physical optics used the wave nature of light to explain phenomenon like interference, diffraction, and superposition. This is the duality of light waves and particle.

Dual nature of matter: On the basis of dual nature of light, in 1924, Louis de Broglie suggested that all the microscopic material particles such as $e^{\ominus}$, protons, atoms, molecules etc.

⇒ If the particle of matter has a momentum then waves associated with it, this is known as de-broglie hypothesis.

⇒ Some sort of waves propagates with moving material particles called matter waves as De-Broglie waves.

⇒ The wavelength associated with matter waves is called de Broglie wavelength.

Its relation is given by $\lambda = \frac{h}{p}$; where $h = 6.6 \times 10^{-34}$ Js

De Broglie wavelength derivation by analogy and its different forms:

A particle of mass 'm' moving with velocity 'c' possess energy given by Einstein's Equation.

$$E = mc^2 \quad \text{---} (1)$$

Acc. to plack's quantum theory the energy of quantum of frequency 'v' is

$$E = h\nu \quad \text{---} (2)$$

from (1) & (2) $mc^2 = h\nu = hc/\lambda$ (since $\nu = c/\lambda$)

$$\lambda = \frac{hc}{mc^2}$$

$$\lambda = h/mc$$

Since $v \approx c$,

$$\therefore \boxed{\lambda = \frac{h}{mv} = \frac{h}{p}} \quad \text{---(1)}$$

where p is the momentum of the particle.

⇒ **de-Broglie wavelength of a free particle in terms of kinetic energy:-**

$$E_K = \frac{1}{2}mv^2$$

multiply × and divide ÷ by m on the right-side term.

$$E_K = \frac{1}{2m} m^2 v^2$$

$$E_K = \frac{p^2}{2m} \quad \text{since } p = mv$$

$$p^2 = 2mE_K$$

$$p = \sqrt{2mE_K} \quad \text{---(2)}$$

Substitute the value of p in de broglie's formula

$$\Rightarrow \boxed{\lambda = \frac{h}{\sqrt{2mE_K}}} \rightarrow (3)$$

⇒ Now, consider a charged particle of charge q, mass m accelerated through a potential difference (voltage) V. This work done on this particle due to the potential difference is given by qV. This work done is converted into the particle's kinetic energy, i.e., $E_K = qV$. Substitute this E_K in an equation (3).

$$\boxed{\lambda = \frac{h}{\sqrt{2mqV}}} \quad \text{---(4)}$$

⇒ If an electron accelerated with potential difference 'V', then the work done on this particle due to the potential difference is 'eV', then eqn (4) becomes,

$$\lambda = \frac{h}{\sqrt{2meV}}$$

After rearranging,

$$\lambda = \frac{h}{\sqrt{2me}} \times \left(\frac{1}{\sqrt{V}}\right) \longrightarrow (4)$$

we know that, Plack's constant: $h = 6.625 \times 10^{-34}$ J S, charge on electron, $e = 1.6 \times 10^{-19}$, mass of electron, $m = 9.1 \times 10^{-31}$ kg

Then eqn (5) becomes,

$$\lambda = \frac{6.625 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}} \times \left(\frac{1}{\sqrt{V}}\right)$$

$$\boxed{\lambda = \frac{12.27}{\sqrt{V}} \text{ Å}} \longrightarrow (5)$$

Matter waves and its properties (characteristics):

1. Phase velocity v_p: The velocity with which a wave travels is called phase velocity and is also called wave velocity.

$$\boxed{v_p = v\lambda = \frac{\omega}{k} = \frac{c^2}{v}} \longrightarrow (6)$$

2. Wave Packet: It is the resultant wave of matter waves which is formed due to the superposition of two or more waves with slightly differ in their frequencies or velocities

⇒ The velocity with which the wave packet propagates is called group velocity which is same as particle velocity.

⇒ It is denoted by v_g

$$\boxed{v_g = \frac{d\omega}{dk}} \longrightarrow (7)$$

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Characteristics (Properties) of matter waves :-

1. Waves associated with moving particles are called matter waves.
2. The wavelength 'λ' of a de-broglie wave associated with particle of mass 'm' moving with velocity 'v' is $\lambda = h/(mv)$.
3. The amplitude of the matter wave depends on the probability of finding the particle in that position.
4. The speed of matter waves depends on the mass and velocity of the particle associated with the wave.
5. The phase velocity has no physical meaning for matter waves.
6. Group velocity is associated with matter waves.

Heisenberg's Uncertainty Principle: It states that "It is impossible to measure simultaneously both the position and momentum of a particle accurately".

⇒ Acc/ to the principle, in any simultaneous determination of position and momentum of the particle, the product of uncertainties is given equal to or greater than $\frac{h}{4\pi}$.

$$\boxed{\Delta x \cdot \Delta p \geq \frac{h}{4\pi}} \longrightarrow (8)$$

where Δx is the error in the measurement of position.
Δp is the error in the measurement of momentum.

Principle of complementarity :-

1. Energy (E) and time (t) with →

$$\hookrightarrow \Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

2. Angular displacement (θ) and angular momentum (L) with

$$\Delta\theta \cdot \Delta L \geq \frac{h}{4\pi}$$

Wave Function: A physical situation in quantum mechanics is represented by a function called wave function.

⇒ It is denoted by ψ.

$$\psi = Ae^{i(kx-\omega t)}$$

Time Independent Schrodiger wave Equation:

consider a particle of mass 'm' moving with velocity 'v'.

The de-Broglie wavelength 'λ' is

$$\lambda = \frac{h}{mv} = \frac{h}{p} \quad \cdots (1)$$

where $p = mv$ is the momentum of the particle.

⇒ The de-Broglie wave equation for a particle travelling in positive x-axis can be written in complex form as

$$\psi = Ae^{i(kx-\omega t)} \quad -(2)$$

where, A is Amplitude.

ω is angular frequency

ψ is the Total wave function and k is the wave no.

⇒ The space independent term in an eq^n (2) can be represented as a wave function,

$$\psi = Ae^{i(kx)} e^{i(-\omega t)}$$

$$\psi = \gamma e^{-i\omega t} \quad -(3)$$

where $\gamma = Ae^{i(kx)}$

Differentiating eq^n (3) w.r.t 'x' twice, we get

$$\frac{d^2\psi}{dx^2} = e^{-i\omega t} \frac{d^2\gamma}{dx^2} \quad -(4)$$

Differentiating eq^n (3) w.r.t 't' twice, we get

$$\frac{d\psi}{dt} = (-i\omega)e^{-i\omega t}\gamma$$

$$\frac{d^2\psi}{dt^2} = (-i\omega)(-i\omega)e^{-i\omega t}\gamma$$

$$\frac{d^2\psi}{dt^2} = (i^2\omega^2)e^{-i\omega t}\gamma$$

$$\frac{d^2\psi}{dt^2} = -\omega^2 e^{-i\omega t}\gamma \quad -(5)$$

we know that, the eqn for a travelling wave in differential form can be written as

$$\frac{d^2y}{dx^2} = \frac{1}{v^2}\frac{d^2y}{dt^2}$$

Then by analogy, the wave eqn for a de-broglie wave for the motion of free particle propagating in x-axis with a velocity v, with a displacement x at a time t, can be written as

$$\frac{d^2\psi}{dx^2} = \frac{1}{v^2}\frac{d^2\psi}{dt^2} \quad \text{— (6)}$$

Substituting eqn (4) and (5) in the above eqn (6), we get

$$e^{-i\omega t}\frac{d^2\psi}{dx^2} = \left(\frac{1}{v^2}\right)(-\omega^2 e^{-i\omega t}\psi)$$

$$\frac{d^2\psi}{dx^2} = -\frac{\omega^2}{v^2}\psi$$

But we know that, $\omega = 2\pi\nu$ and $v = \nu\lambda$. Substituting these values in the above eqn

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2\nu^2}{\nu^2\lambda^2}\psi = \frac{-4\pi^2}{\lambda^2}\psi \quad \text{— (7)}$$

from eqn (1) substitute for λ in the above eqn

$$\frac{d^2\psi}{dx^2} = -\frac{4\pi^2 m^2 v^2}{h^2}\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2}\psi = 0 \quad \text{— (8)}$$

we know that, Total energy = kinetic energy + Potential energy

$$E = \frac{1}{2}mv^2 + V$$

multiply both sides by 2m, we get

$$2mE = m^2v^2 + 2mV$$

$$m^2v^2 = 2m(E - V) \quad \text{— (9)}$$

Substitute eqn (9) in the eqn (8), then we get

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2(2m)\cdot(E-V)}{h^2}\psi = 0$$

$$\boxed{\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m(E-V)}{h^2}\psi = 0}$$

This is known as time independent Schrondinger wave eqn in 1-D

$$\boxed{\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2 m(E-V)}{h^2}\psi = 0}$$

This is known as time independent Schrodinger wave eqn in 3-D.

Electrical conductivity in metals

Ques-1- Enumerate the failures of classical free eΘ theory and discuss the assumption of quantum free eΘ theory.

Ans- 01- Specific heat capacity:- The theory predicted a specific heat capacity for metals (4.5R, where R is the universal gas constant) that significantly differed from the experimentally observed value approximately 3R) based on Dulong petit's law.

⇒ This discrepancy highlights a fundamental flow in the theory's handling of the energy contribution of free electrons to the total heat capacity.

⇒ Temperature Dependence of Electrical conductivity:- The theory predicts that electrical conductivity decreases with increasing temperature due to increased scattering of eΘ by lattice vibrations. However, it fails to accurately account for the complex interplay of factors that determine conductivity, particularly in semiconductors and insulators.

$$\sigma \propto \frac{1}{\sqrt{T}}$$

σ = conductivity

T = Temperature

⇒ Dependence of Temperature of Electrical conductivity on Electron concentration.

Ans- From the expression for electrical conductivity $\sigma = ne^2\tau/m$ conductivity is directly proportional to the electron density n.

⇒ Hence, divalent and trivalent metals should passes much higher electrical conductivity than monovalent metals.

⇒ This is contray to the experimental observation that silver and copper (monovalent metals) are more conductivity than zinc and aluminium (divalent and trivalent metals.

⇒ Assumption of quantum free electrons theory:-

⇒ The valence eΘ are free to move about inside the metal. But they cannot escape from the metal due to

the potential barrier at the surface.
⇒ The force of attraction between the free $e^{\ominus}$ and the ionic core and the force of repulsion b/w $e^{\ominus}$ is negligible.
⇒ The potential inside the metal is constant.
⇒ The energies of $e^{\ominus}$ are quantized.
⇒ They allowed discrete energy levels obey pauli's exclusion principle, according to which no two electrons in an energy level can have the same set of quantum numbers.

Ques-2. Discuss the concept of phonon and explain mattheissen's law:

Ans- Concept of phonon:- The atoms or ions in the lattice are in constant motion, vibrating about their equilibrium positions due to thermal energy. These vibrations give rise to what we call as phonons.
⇒ Phonons represent the vibrational modes of atoms or ions in a crystal lattice.
⇒ Phonons are quatized units of lattice vibrations, similar to the how photons are quantized unit of light. They carry both energy and momentum.
⇒ When an $e^{\ominus}$ moves through the crystal lattice it can intract with phonons.
⇒ These interaction can scatter electrons, leading to phenomena like electrical resistance and heat conductivity.

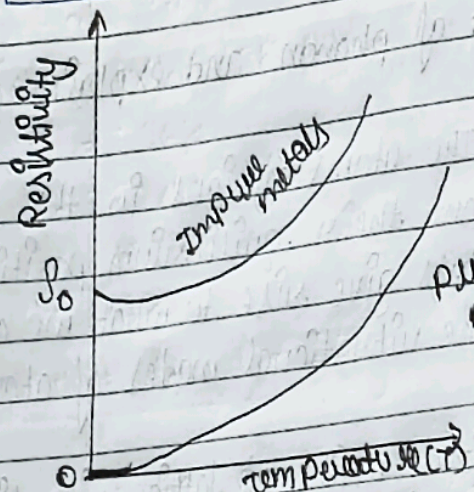
Mattheissen's Rule: This is a principle that describes the total electrical resistivity of a crystalline material as the sum of its individual resistance due to various scattering mechanisms.
⇒ Mattheissen's rule states that the total electrical resistivity associa-ed with each of these scattering processes. Mathematically, it can be expressed as:

$\rho_{total} = \rho_{impurities} + \rho_{phonons} + \rho_{electron\ electron\ interaction}$

⇒ Each term on the right side represents the contribution to the resistivity due to a specific scattering mechanism.

⇒ By considering these individual contribution, Matthiessen's rule provides a way to estimate the total resistivity of a material.
⇒ The two kinds of contributions towards the total resistivity: a temperature-independent resistivity (ρ_0) due to impurities and temperature dependent resistivity which is the sum of all other resistivities.
⇒ So, we can write mathematics matheissen's rule simply as:

$$\rho_{total} = \rho_0 + \rho_T$$

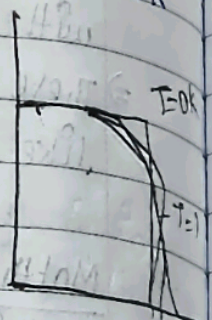


Ques-③ Define fermi energy. Discuss various energy states by the $e^{\ominus}$ at $T=0^{\circ}K$ and $T>0^{\circ}K$ on the basis of fermi factors.

Ans- The highest energy level than an electron can passes at 0K in a metal is called Fermi energy.

Case 1: $f(E)$ for $E<E_F$ at $T=0K$

when $E<E_F$ and $T=0K$, $\frac{E-E_F}{k_BT}$ becomes $\to \infty$



Therefore $f(E) = \frac{1}{1+e^{\frac{E-E_F}{k_BT}}}$ becomes

$$f(E<E_F)\,/\,T=0K = \frac{1}{1+e^{-\infty}} = 1$$

because $e^{-\infty} = 0$

This means that at $T=0K$, none of the energy levels above the fermi level are occupied by $e^{\ominus}$.
There is 100% prabability of finding an $e^{\ominus}$ below the fermi

energy level at absolute zero temperature.

⇒ Case 02: $f(E)$ for if $E > E_F$ at $T = 0K$

when $E > E_F$ and $T = 0K$, $\frac{E - E_F}{K_B T}$ becomes ∞

Then $f(E > E_F) / T = 0K = \frac{1}{1 + e^{\infty}} = 0$

because $e^{\infty} = \infty$

This means that at $T = 0K$, none of the energy levels above the fermi level are occupied by electrons.

There is 0% probability of finding an electron above the fermi energy level at absolute zero temperature.

⇒ Case 03: $f(E)$ for $E = E_F$ at $T > 0K$

when $E = E_F$ and $T > 0K$, $\frac{E - E_F}{K_B T}$ becomes 0

Therefore,

$$f(E - E_F) / T > 0K = \frac{1}{1 + e^0} = \frac{1}{2}$$

because $e^0 = 1$.

This means that at $T > 0K$, there is 50% probability of finding an electron at the fermi energy level.

Ques-4 write about Fermi-Dirac statistics.

Ans - ⇒ At temperature above 0K, one cannot tell with certain whether a particular energy level will be occupied by an electron or not.

⇒ For example, under thermal equilibrium, the free electrons occupy varies energy levels in accordance with a statistical rule knows as fermi-Dirac statistics.

⇒ Fermi-Dirac statistics enables the evaluation of probability of finding electron in energy levels over a certain range of energy values.

⇒ The evaluation is done with the help of a quantity called

Fermi factor $f(E)$ is given by

$$f(E) = \frac{1}{1 + e^{\frac{E-E_F}{KBT}}}$$

⇒ Fermi factor is the probability of occupation of a given energy state for a material at thermal equilibrium.

Problem-

Ques- let us superconducting transion temperature of 7.26K if the initial field at 0°K is 50×10^3 A/m calculate the critical field at 6K.

Soln- Given:- $T_c = 7.26K$

$H_0 = 50 \times 10^3$ A/m

$H_c = ?$

$T = 6K$

$$H_c(T) = H_0 \left(1 - \frac{T^2}{T_c^2}\right)$$

$$H_c(T) = 50 \times 10^3 \left(1 - \frac{6^2}{7.26^2}\right)$$

$$H_c(T) = 50 \times 10^3 \left(1 - \frac{36}{52.7076}\right)$$

$$H_c(T) = 15849.32 \text{ A/m}$$

ues- Superconducting tin has critical magnetic field of $B_0 =$ 0.0217T at 2K. If the critical temperature for super conducting transition is 3.7K, find the critical magnetic field at 3K?

Hⁿ- Given:- $T_c = 3.7K$

$B_{c_0} = 0.0217T$

$T_1 = 2K$

$B_c = ?$

$T_2 = 3K$

As formula, $B_c(T) = B_0\left(1 - \frac{T^2}{T_c^2}\right)$

$$B_{c1} = B_0\left(1 - \frac{T_1^2}{T_c^2}\right) \quad \text{— (1)}$$

and $$B_{c2} = B_0\left(1 - \frac{T_2^2}{T_c^2}\right) \quad \text{— (2)}$$

then Ratio $$\frac{B_{c2}}{B_{c1}} = \frac{\left(1 - \frac{T_2^2}{T_c^2}\right)}{\left(1 - \frac{T_1^2}{T_c^2}\right)}$$

then, $$B_{c2} = B_{c1}\frac{\left[1 - \frac{T_2^2}{T_c^2}\right]}{\left[1 - \frac{T_1^2}{T_c^2}\right]}$$

$$B_{c2} = 0.0217\frac{\left[1 - \frac{2^2}{(3.7)^2}\right]}{\left[1 - \frac{(2)^2}{(3.7)^2}\right]}$$

$$B_{c2} = 1.545 \times 10^{-3}$$

Ques- A superconducting tin has critical temp of 3.7K at $0\vec{B}$. And a critical field of 0.0306T at 0K. Find the critical field at 2K.

Soln Given: $B_1 = 0.0306T$

$T_1 = 0K$

$B_c =$

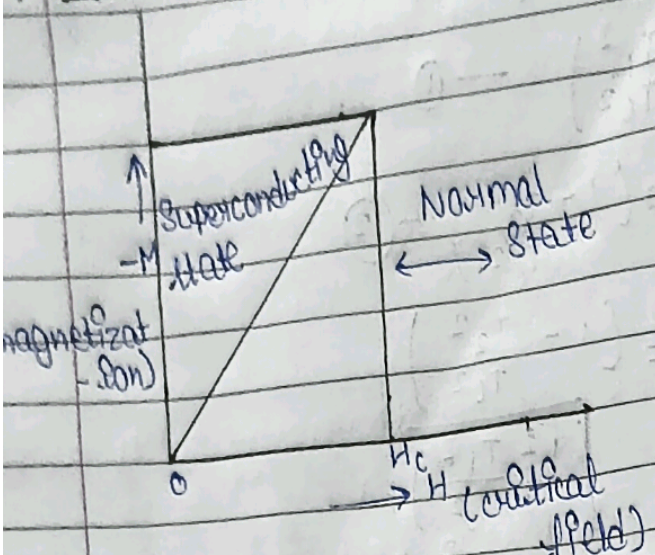
$T_2 = 2K$

$T_c = 3.7K$

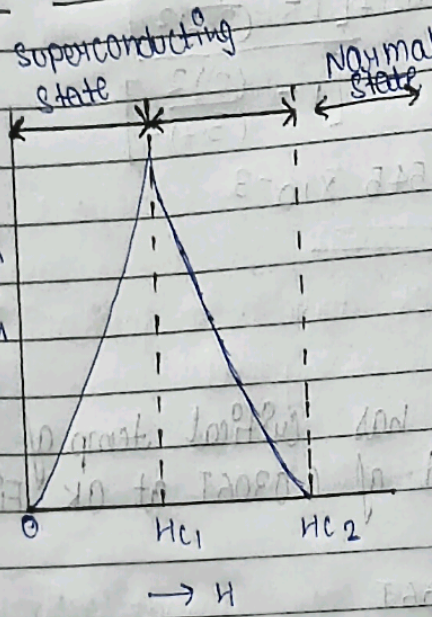
$$B_{c2} = 0.0306\left(\frac{1 - \frac{0^2}{(3.7)^2}}{1 - \frac{2^2}{(3.7)^2}}\right)$$

$$B_{c2} = 0.0216T$$

⇒ Type - I Semiconductor - It undergoes completely in Meissner effect



⇒ Type - II Semiconductor -



⇒ BCS Theory → Barden - Copper - Schriffier theory :

⇒ Combination of all charges.

⇒ Phonon is vibration of charges at lattice point.